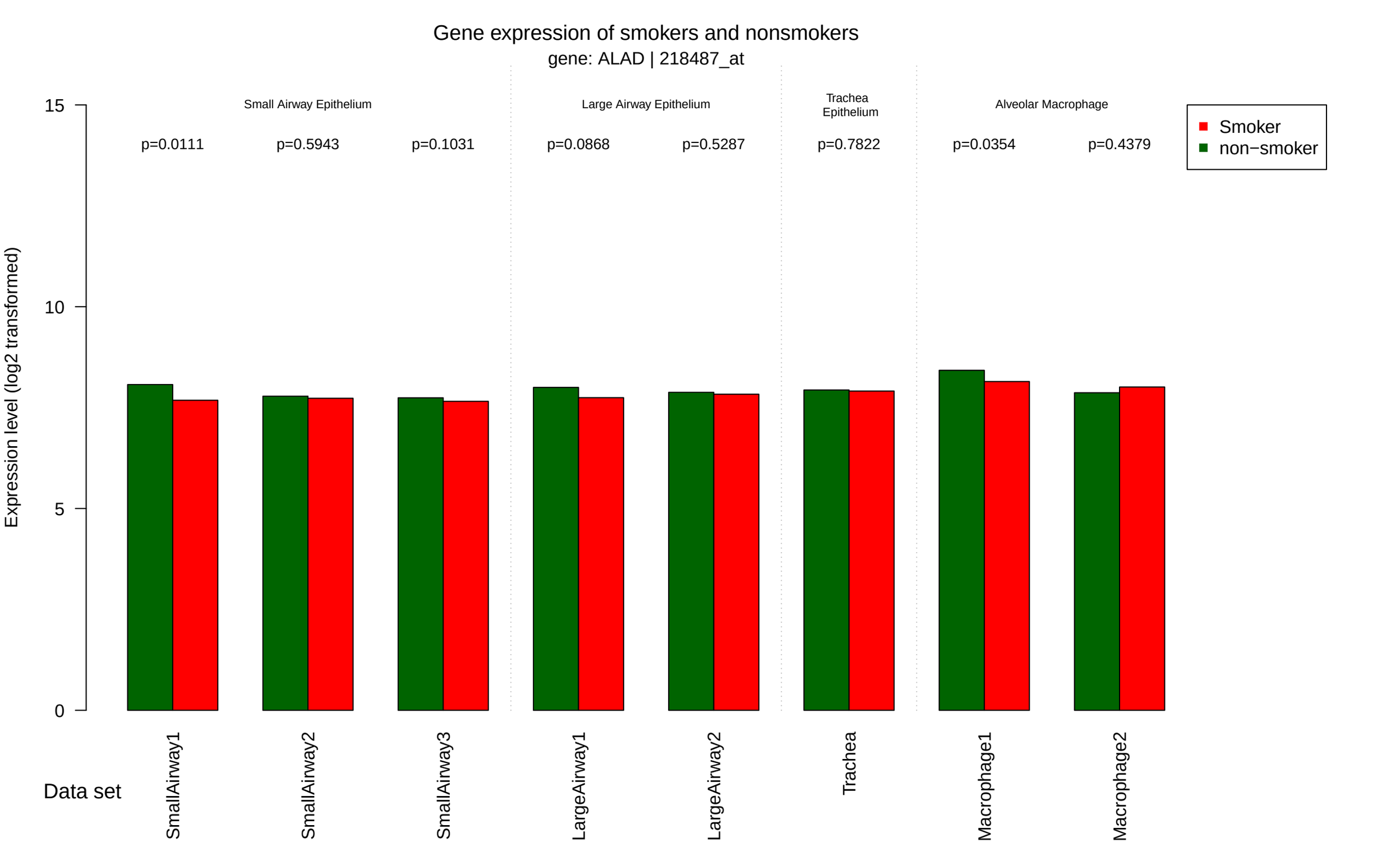
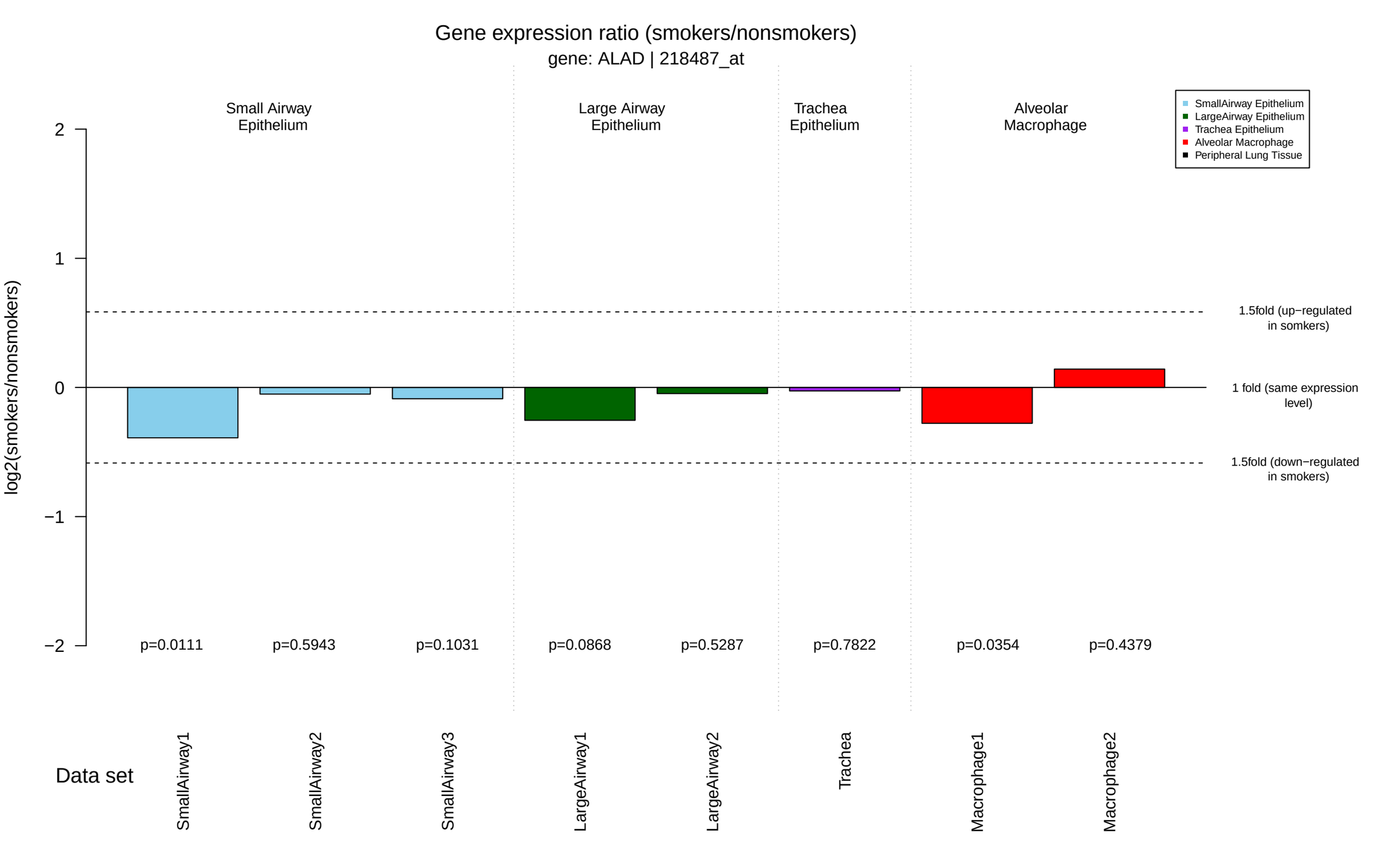
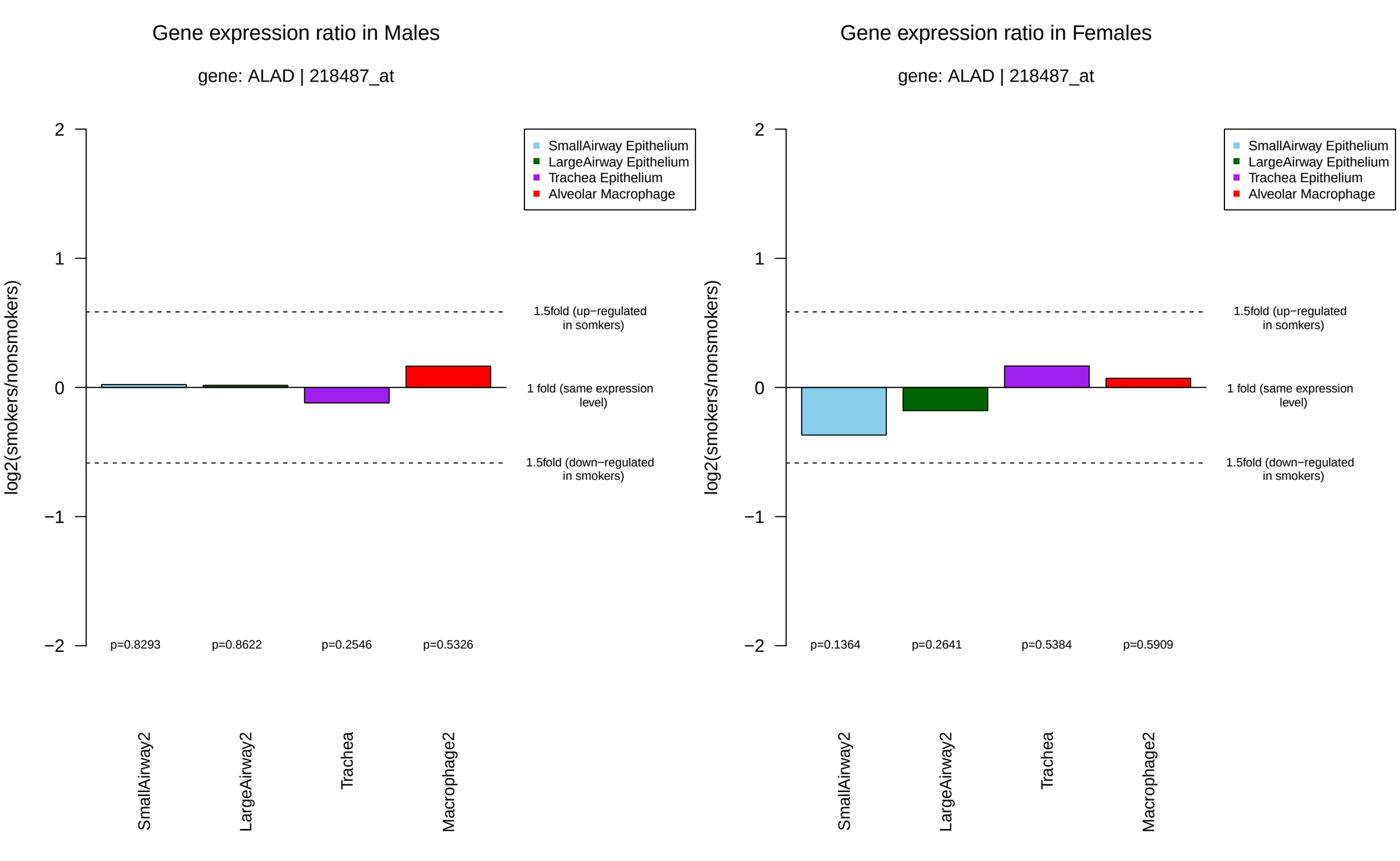
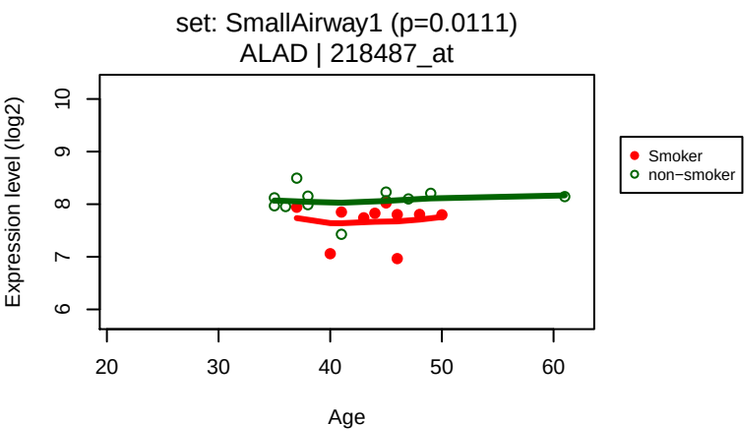
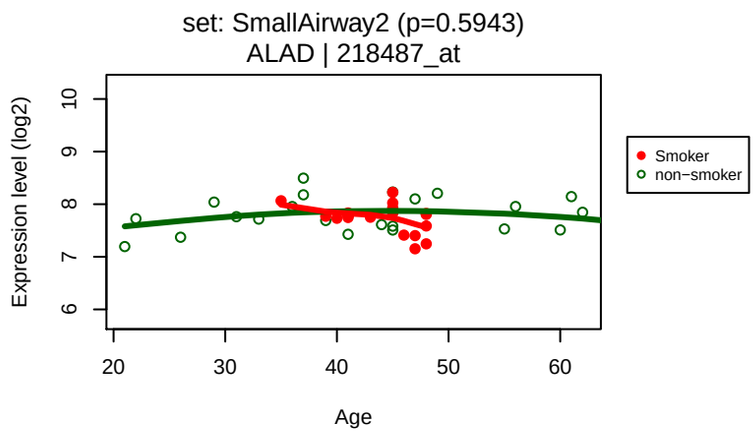
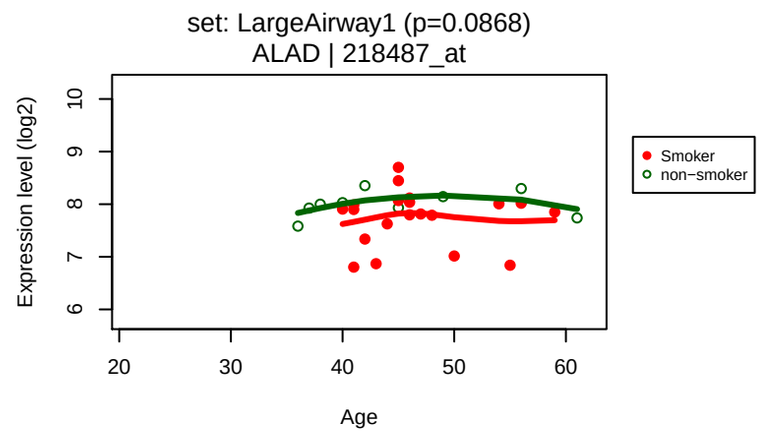

Gene expression of smokers and nonsmokers
gene: ALAD | 218487_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0111
p=0.5943
p=0.1031
p=0.0868
p=0.5287
p=0.7822
p=0.0354
p=0.4379
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

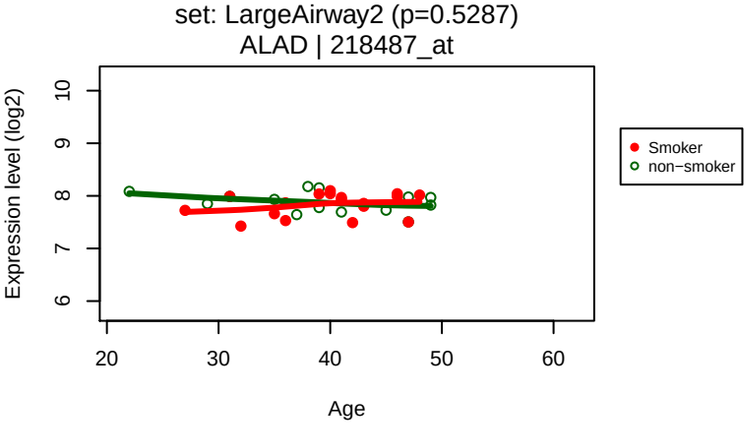

Gene expression ratio (smokers/nonsmokers)
gene: ALAD | 218487_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0111
p=0.5943
p=0.1031
p=0.0868
p=0.5287
p=0.7822
p=0.0354
p=0.4379
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

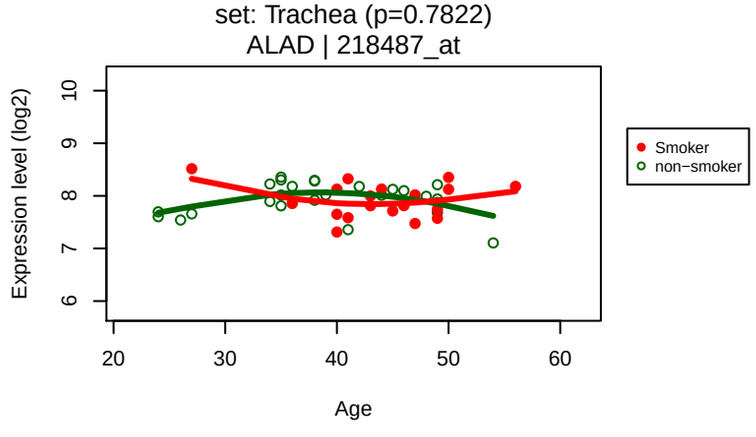

Gene expression ratio in Males
gene: ALAD | 218487_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8293
p=0.8622
p=0.2546
p=0.5326
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALAD | 218487_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1364
p=0.2641
p=0.5384
p=0.5909
SmallAirway2
LargeAirway2
Trachea
Macrophage2

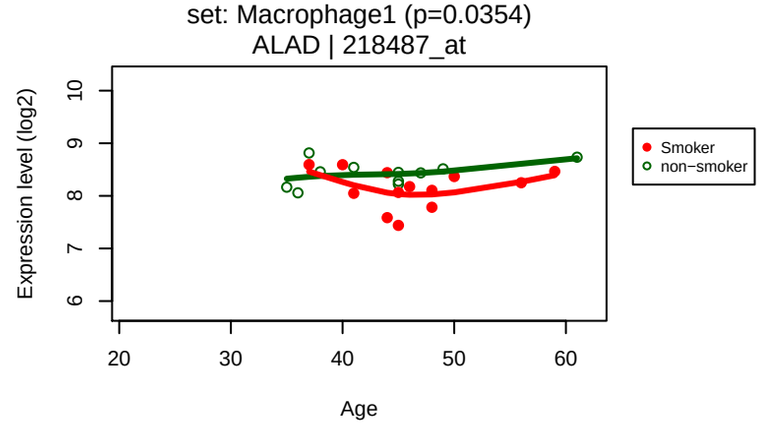

set: SmallAirway1 (p=0.0111)
ALAD | 218487_at
Expression level (log2)
Age
Smoker
non−smoker

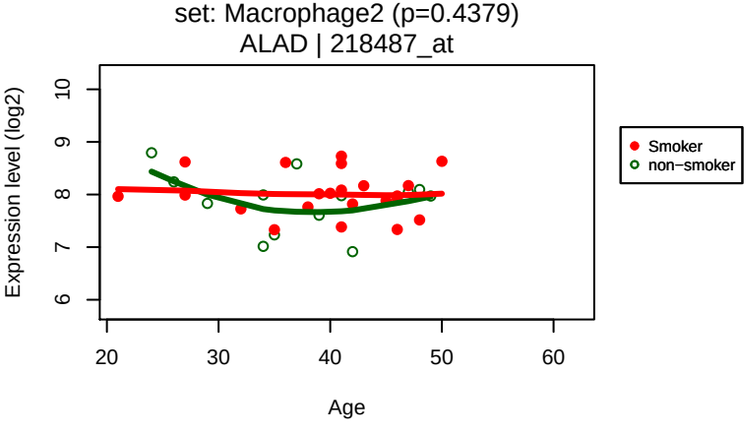

set: SmallAirway2 (p=0.5943)
ALAD | 218487_at
Expression level (log2)
Age
Smoker
non−smoker

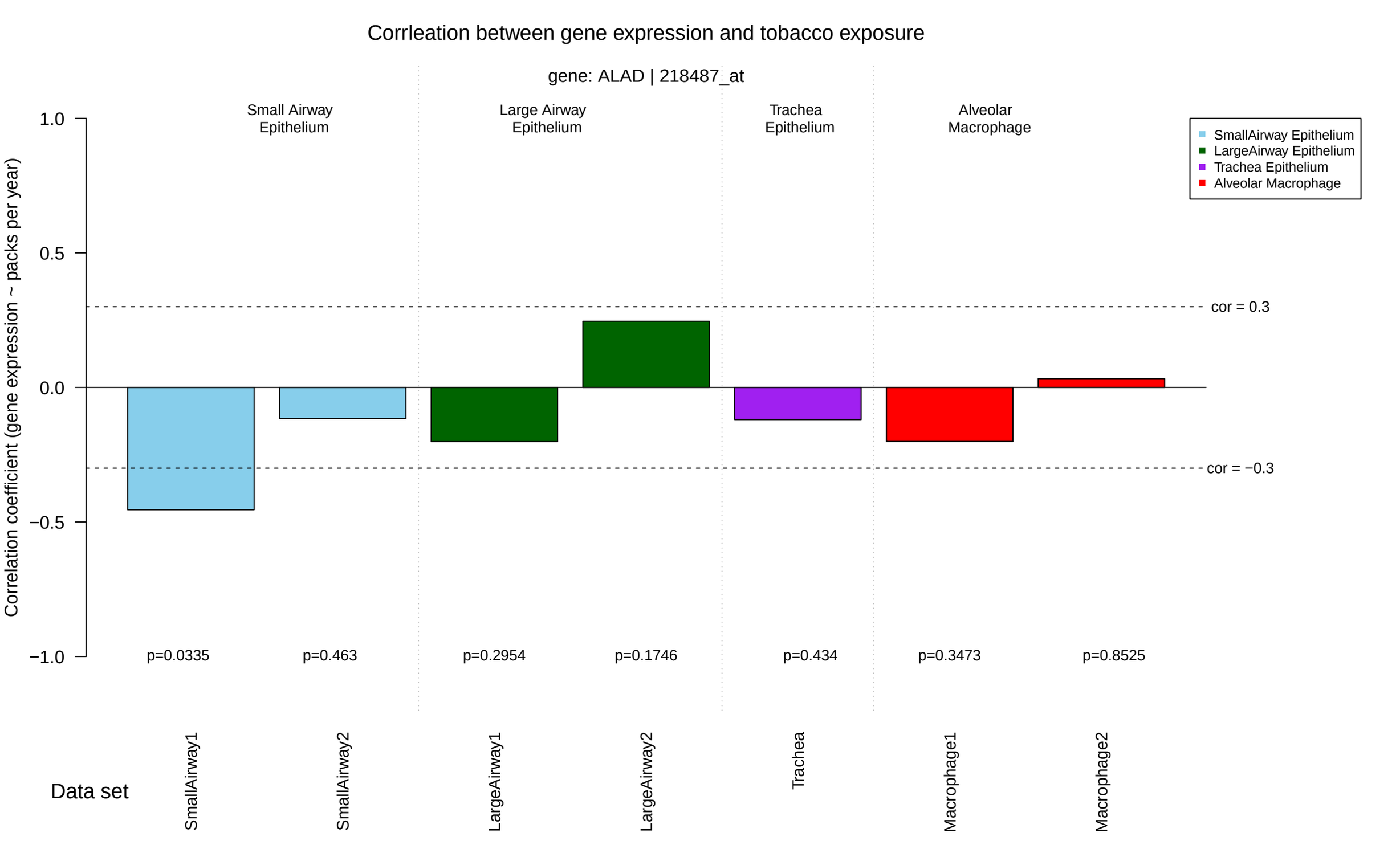

set: LargeAirway1 (p=0.0868)
ALAD | 218487_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5287)
ALAD | 218487_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7822)
ALAD | 218487_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0354)
ALAD | 218487_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4379)
ALAD | 218487_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALAD | 218487_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0335
p=0.463
p=0.2954
p=0.1746
p=0.434
p=0.3473
p=0.8525
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2